

HDF5 Audit Report

[NN]

Very abstract...

hello

Acknowledgments: This material is based upon work supported by the U.S. National Science Foundation under Federal Award No. 2534078. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

Revision History

Version Number	Date	Comments
v1	July 4, 2026	Initial draft released for internal review.
v2	August 1, 2026	Public release.

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1 Introduction

1.1 Purpose

1.2 Scope

Out of scope:

- PHDF5
- HDFView
- HSDS
- AI

1.2.1 Safety

1.2.2 Security

1.2.3 Privacy

2 Motivation

2.1 Problem Statement

2.2 Goals

3 HDF5 Core Library and File Format

3.1 Accidents Waiting to Happen – HDF5 Safety

- Disk full
- File corruption
- Long term data accessibility and interpretability

3.2 Verify and Verify – HDF5 Security

- Integer arithmetic
- Running with scissors

3.3 Footprints, Fingerprints, Controls – HDF5 Privacy

- It's all in the open
- Middle ground
- All-out encryption

3.4 HDF5 Tools

3.5 Audit Findings and Mitigation Priorities

4 HDF5 Extensions

4.1 Overview

4.2 Virtual Object Layer (VOL) Connectors

4.3 Filters Plugins

4.4 Virtual File Drivers (VFD)

4.5 Audit Findings and Mitigation Priorities

5 HDF5 Wrappers and Language Bindings

5.1 Overview

Complete:

- C#
- Java
- Python
- JavaScript
- Rust

Partials...

5.2 Python

- CVE-2025-995 [8]
- CVE-2026-0897 [9]
- CVE-2026-1669 [10]
- CVE-2026-2492 [7]

5.3 Java

5.4 Audit Findings and Mitigation Priorities

6 Independent HDF5 Implementations

6.1 Overview

- HDF5 library playing footsie with the file format spec: GitHub issue 6409 [\[1\]](#)

6.2 Audit Findings and Mitigation Priorities

7 HDF5 in the Software Supply Chain

- HDF5 2.0.0 source-artifact integrity check [\[11\]](#)
- SBOM

7.1 Audit Findings and Mitigation Priorities

8 Long-term Data Accessibility and Interpretability

8.1 Audit Findings and Mitigation Priorities

9 Resources

- HDF SSP SIG [6]
- Tools
 - HDF5 CVE test suite [3]
 - A machine-readable specification of the HDF5 file format. [2]
 - HDF5 extension templates [4]
 - HDF5 plugins [5]
- S2D2

References

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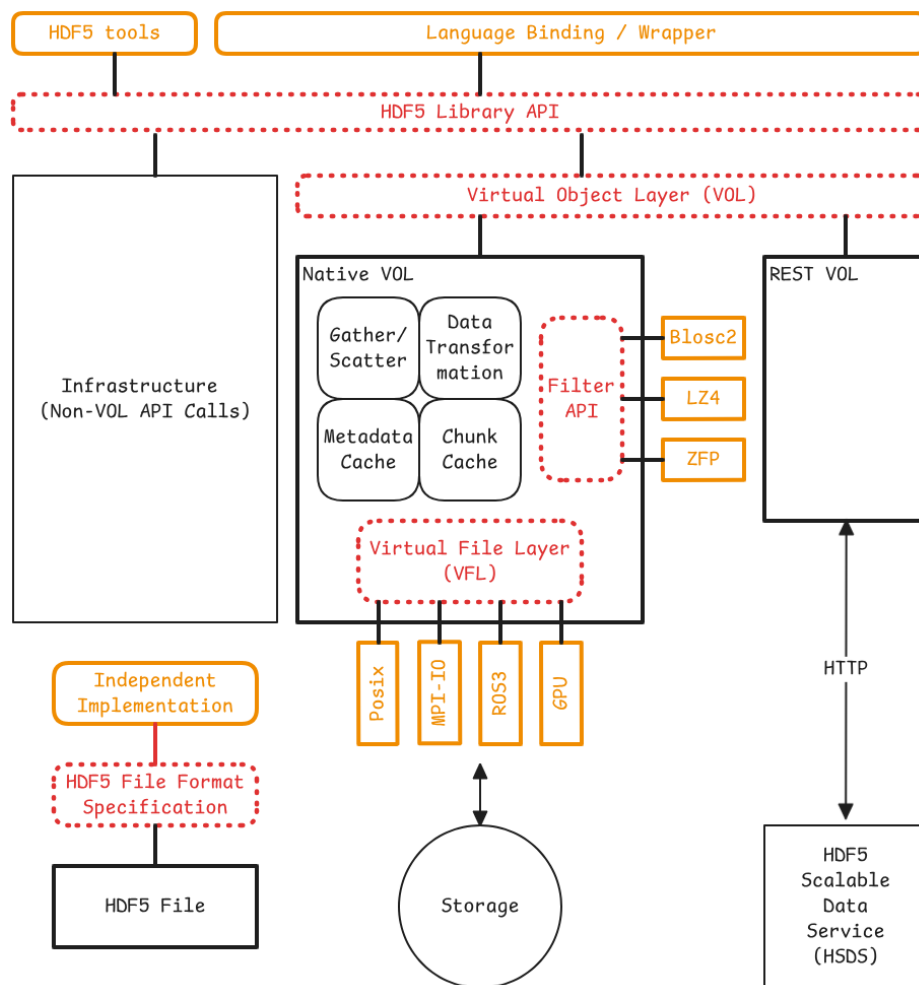


Figure 1 – HDF5 vulnerability surfaces (APIs: red – Supply chain: orange)